

Neutrino trident production at MicroBooNE and the Short-Baseline Neutrino Program: Standard Model predictions and probes of dark neutrino models and on-shell Z' emission

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We revisit the rate of neutrino-induced trident production $\nu_\ell A \rightarrow \nu_\ell \ell'^+ \ell'^- A$ at liquid-argon detectors exposed to the Booster Neutrino Beam (BNB) and the off-axis NuMI flux, focusing on the dimuon final state at the Short-Baseline Neutrino (SBN) detectors – SBND, MicroBooNE, and ICARUS. Using a dedicated event generator, NEPTUNE, we compute coherent, diffractive, and incoherent contributions on argon at the ~ 1 GeV energies relevant to SBN, and quantify uncertainties from nuclear form factors and from the matching between low- and high- Q^2 regimes. We then evaluate two beyond-the-Standard-Model enhancements that yield identical $\mu^+ \mu^-$ topologies: (i) cascade decays of a heavy neutral lepton (HNL) produced in coherent up-scattering, as expected in dark neutrino models, and (ii) on-shell emission of a kinetically mixed or $L_\mu - L_\tau$ -like dark gauge boson Z' through the neutrino-induced inverse Primakoff process, $\nu A \rightarrow \nu Z' A$, followed by $Z' \rightarrow \mu^+ \mu^-$. We compare expected SBN event rates and differential distributions across these scenarios. [The Results and Conclusions sections are reserved for the analysis of NEPTUNE output.]

I. INTRODUCTION

Neutrino trident production – the radiative emission of a charged lepton pair off the electroweak current of a neutrino traversing the Coulomb field of a heavy nucleus,

$$\nu_\ell A \longrightarrow \nu_\ell \ell_1^+ \ell_2^- A, \quad (1)$$

is one of the rarest electroweak processes accessible in present and near-future neutrino experiments. Its leading-order rate is suppressed by four powers of the weak coupling and by an additional factor of α^2 relative to ordinary charged-current scattering, and it features a characteristic dilepton topology with very little hadronic activity, making it both an exquisite test of the Standard Model (SM) and a sensitive probe of new physics that couples to leptons [1–6].

The process was first studied theoretically in the 1960s [7–10] in the equivalent-photon approximation (EPA), in which the nucleus is modeled as a source of quasi-real photons that fuse with the neutrino weak current to produce the lepton pair. The first experimental observation came from CHARM-II [11], followed by measurements at CCFR [12] and NuTeV [13], all in the dimuon channel and at neutrino energies $E_\nu \gtrsim 20$ GeV. Modern liquid-argon time-projection chambers (LArTPCs) operate at much lower energies, where the EPA is at best marginal: ArgoNeuT recently reported a measurement of dimuon trident production on argon at neutrino energies of a few GeV, observing $0.13^{+0.09}_{-0.07}$ events/ton/10²⁰ POT in the NuMI on-axis flux [14], in agreement with the SM prediction within sizable statistical uncertainties. MicroBooNE has since searched for $\nu_\mu A \rightarrow \nu_\mu \mu^+ \mu^- A$ in the BNB

and NuMI fluxes [15], and the upcoming Short-Baseline Neutrino (SBN) program at Fermilab [16] – comprising SBND (112 ton active mass), MicroBooNE (85 ton), and ICARUS (476 ton) – is expected to collect samples of trident events large enough to challenge precision predictions and to set competitive limits on lepton-specific extensions of the SM.

Two developments motivate a dedicated reanalysis of trident production at SBN energies. First, the EPA breaks down in the kinematic regime $E_\nu \sim \text{few GeV}$ and $Q^2 \lesssim m_\mu^2$, and a fully 2 $\rightarrow$ 4 matrix-element treatment is required to capture the interference between coherent, diffractive, and inelastic contributions, the helicity structure of the leptonic final state, and the smearing induced by the nuclear form factor of ⁴⁰Ar [3, 5, 6]. Second, light vector mediators with sub-GeV masses – including gauged $L_\mu - L_\tau$ bosons [17–19], dark photons [20, 21], and more general dark gauge bosons that arise in dark neutrino models [22–25] – enter the trident amplitude either as s -channel resonances in the dilepton invariant mass or, more interestingly, as on-shell final-state particles in association with the outgoing neutrino. The latter mechanism, $\nu A \rightarrow \nu Z' A$ with $Z' \rightarrow \mu^+ \mu^-$, is closely analogous to the Primakoff production of axion-like particles in nuclear Coulomb fields [26] and provides a complementary probe of light-mediator parameter space relative to existing constraints from $g_\mu - 2$ [27–29], $B \rightarrow K \nu \bar{\nu}$ [30], and beam-dump searches [31, 32]. The compositional overlap of trident events with Z' -enhanced channels has motivated several recent proposals to use SBN, DUNE, and FASER-type detectors to discriminate among models [4, 33–35].

In this work we provide updated predictions for SM trident production on argon under the BNB and the off-axis NuMI fluxes incident at SBND, MicroBooNE, and ICARUS, and we map the parameter space of two representative BSM scenarios that yield additional $\mu^+ \mu^-$

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events indistinguishable on a topological basis from SM trident. All rates are computed with NEPTUNE, a Monte Carlo generator developed for neutrino-induced rare processes that retains the full $2 \rightarrow n$ matrix element including nuclear form factors and lepton mass effects. The two BSM benchmarks are: (i) coherent up-scattering of an active neutrino into a heavy neutral lepton (HNL), $\nu_\alpha A \rightarrow N A$, followed by $N \rightarrow \nu \mu^+ \mu^-$ via an off-shell or on-shell Z' – the production mechanism originally introduced to address the MiniBooNE low-energy excess [22–25]; and (ii) the neutrino-induced inverse Primakoff process, $\nu_\mu A \rightarrow \nu_\mu Z' A$ with on-shell $Z' \rightarrow \mu^+ \mu^-$, in L_μ – L_τ and kinetically mixed dark-photon models.

The remainder of this paper is organized as follows. In Sec. II we review trident production in the SM, detail our matrix-element treatment, and discuss the partition into coherent, diffractive, and incoherent regimes at SBN energies. In Sec. III we describe the dark neutrino model and its trident-like signature, and in Sec. IV we formulate the on-shell Z' production process as a neutrino-induced inverse Primakoff reaction. Section V summarizes NEPTUNE and the input fluxes and detector parameters used for SBN. Sections VI and VII are reserved for the numerical results and conclusions.

II. TRIDENT PRODUCTION IN THE STANDARD MODEL

A. Effective Lagrangian and matrix element

At neutrino energies relevant to SBN, $E_\nu \lesssim 10$ GeV, the momentum transfer to the leptonic system is far below the electroweak scale, $|q^2| \ll M_W^2, M_Z^2$, and the four-fermion limit of the SM applies. After integrating out the massive electroweak bosons and performing a Fierz reordering of the W -exchange contribution, the leptonic interaction relevant for $\nu_\ell \gamma^* \rightarrow \nu_\ell \ell'^+ \ell'^-$ takes the form

$$\mathcal{L}_{\text{eff}} = -\frac{G_F}{\sqrt{2}} [\bar{\nu}_\ell \gamma^\mu (1-\gamma_5) \nu_\ell] [\bar{\ell}' \gamma_\mu (C_V^{\ell\ell'} - C_A^{\ell\ell'} \gamma_5) \ell'], \quad (2)$$

with effective vector and axial couplings

$$C_V^{\ell\ell'} = 2 \sin^2 \theta_W - \frac{1}{2} + \delta_{\ell\ell'}, \quad (3)$$

$$C_A^{\ell\ell'} = -\frac{1}{2} + \delta_{\ell\ell'}. \quad (4)$$

The Kronecker delta encodes the additional W -exchange contribution when the trident lepton flavor matches the incoming neutrino flavor; in particular, for the SBN-relevant channel $\nu_\mu A \rightarrow \nu_\mu \mu^+ \mu^- A$ both W and Z contribute and interfere constructively in the vector coupling.

The full $2 \rightarrow 4$ amplitude on a nucleus of charge Z and mass number A is obtained by attaching the leptonic tensor of Eq. (2) to a virtual photon emitted by the nucleus,

with the appropriate electromagnetic form factor:

$$\mathcal{M} = \frac{e^2 G_F}{\sqrt{2} q^2} F_A(q^2) J_\mu^{\text{had}}(P, P') L_{\text{trid}}^\mu(p_\nu, p_{\nu'}, p_+, p_-), \quad (5)$$

where $q = P - P'$ is the momentum transferred to the nucleus and $F_A(q^2)$ is the nuclear electromagnetic form factor (parametrized below as a Helm or symmetrized Fermi form factor for ^{40}Ar). The leptonic current L_{trid}^μ is computed from the seven tree-level diagrams obtained by attaching the photon to either of the two outgoing charged leptons, summed coherently across W - and Z -exchange topologies, and including the lepton mass throughout. We retain all kinematic dependence: the EPA limit, in which L^μ is replaced by $\sigma(\nu\gamma \rightarrow \nu\ell^+\ell^-)$ folded with an equivalent-photon flux, is recovered for $E_\nu \gg m_\mu$ and small q^2 but is known to overestimate the cross section by tens of percent in the SBN regime [3, 5].

B. Coherent, diffractive, and incoherent regimes

The total trident cross section on a nuclear target receives three distinct contributions, separated by the typical four-momentum transfer to the hadronic system:

1. *Coherent* scattering off the entire nucleus, dominant for $|q^2| \lesssim 1/R_A^2 \sim (60 \text{ MeV})^2$ where R_A is the nuclear radius. This regime is enhanced by a factor of Z^2 from the coherent sum over protons and is described by the elastic nuclear form factor $F_A(q^2)$.
2. *Diffractive* (quasielastic) scattering off individual protons, which dominates for $1/R_A^2 \lesssim |q^2| \lesssim 1/r_p^2$, where $r_p \simeq 0.84$ fm is the proton charge radius. Here the cross section is governed by the proton electric and magnetic form factors $G_E^p(q^2)$ and $G_M^p(q^2)$.
3. *Inelastic / deep-inelastic* scattering off quarks for $|q^2| \gtrsim 1 \text{ GeV}^2$, described by the photon parton distribution function of the nucleon.

At BNB peak energies ($E_\nu \approx 0.8$ GeV) the coherent component is responsible for $\sim 95\%$ of the dimuon trident rate on argon [3, 5]; the diffractive component becomes appreciable in the higher-energy NuMI tail, while the DIS contribution remains negligible across the SBN energy range. We implement all three regimes in NEPTUNE and apply a smooth matching prescription at $|q^2| \simeq 1/r_p^2$ to avoid double counting, following Ref. [5].

C. Phenomenology at ~ 1 GeV

The threshold for dimuon trident production is $\sqrt{s_{\text{min}}} = 2m_\mu \simeq 211$ MeV in the $\nu\gamma$ subsystem, but the cross section rises only logarithmically with E_ν , so SBN

Placeholder for Fig. 1.
Representative tree-level Feynman diagrams for the processes considered in this work.

FIG. 1. Representative tree-level diagrams. Top row: SM trident production through W -exchange and Z -exchange in the coherent regime. Middle row: BSM contribution from an s -channel Z' in $L_\mu - L_\tau$ models, and HNL-mediated cascade in dark neutrino models, $\nu A \rightarrow NA \rightarrow (\nu Z')A \rightarrow (\nu \mu^+ \mu^-)A$. Bottom row: on-shell Z' emission via the neutrino-induced inverse Primakoff reaction, $\nu A \rightarrow \nu Z' A$ followed by $Z' \rightarrow \mu^+ \mu^-$, with both the $\nu \nu Z'$ vertex (e.g. $L_\mu - L_\tau$) and the $\gamma - Z'$ kinetic-mixing topology shown.

Placeholder for Fig. 2.

FIG. 2. Energy spectra of ν_μ and $\bar{\nu}_\mu$ entering SBND, MicroBooNE, and ICARUS from the on-axis BNB and the off-axis NuMI flux, normalized per proton on target. Curves are extracted from the official SBN flux files [16] and rebinned for the NEPTUNE event generation.

sits in the deep-threshold regime. In the coherent EPA, the leading-log expression

$$\sigma_{\nu_\ell A \rightarrow \nu_\ell \mu^+ \mu^- A}^{\text{coh}} \sim \frac{2 G_F^2 \alpha^2 Z^2}{9\pi^3} (C_V^2 + C_A^2) E_\nu^2 \ln^2\left(\frac{E_\nu^2}{m_\mu^2}\right), \quad (6)$$

captures the qualitative scaling but overestimates the rate by $\mathcal{O}(30\%)$ at $E_\nu = 1$ GeV once the lepton masses, the Q^2 dependence of the trident vertex, and the nuclear form factor are properly retained. The angular distribution of the dimuon system is sharply forward, with typical opening angles $\theta_{\mu\mu} \sim m_\mu/E_\nu$, and the leading muon carries a fraction 0.4–0.7 of the available energy. These topological features – small dimuon invariant mass, forward boost, low hadronic activity – are essential to discriminate trident from charged-current single-pion backgrounds at SBN, and also to identify the BSM modifications discussed below.

III. BEYOND THE STANDARD MODEL: DARK NEUTRINO MODELS

A. Model setup

We consider an extension of the SM by a single Dirac heavy neutral lepton N of mass m_N and a new Abelian gauge boson Z' of mass $m_{Z'}$, both odd under a global lepton symmetry that mixes softly with the SM [22–25]. The active–HNL mixing is parametrized by the elements

Placeholder for Fig. 3.

FIG. 3. SM trident cross section $\sigma(\nu_\mu A \rightarrow \nu_\mu \mu^+ \mu^- A)$ on argon as a function of the incoming neutrino energy, decomposed into coherent, diffractive, and DIS contributions as computed by NEPTUNE. The EPA leading-log estimate of Eq. (6) is shown for comparison. The shaded band reflects the uncertainty from the nuclear form factor and from the matching to the diffractive regime.

$U_{\alpha 4}$ of the extended PMNS matrix,

$$\nu_\alpha = \sum_{i=1}^3 U_{\alpha i} \nu_i + U_{\mu 4} N_R^c + \dots, \quad \alpha = e, \mu, \tau, \quad (7)$$

and the gauge interaction couples Z' to a vector current that, in the broken phase, includes off-diagonal ν – N entries:

$$\mathcal{L}_{Z'} \supset g_D Z'_\mu [\bar{N} \gamma^\mu N + U_{\mu 4} \bar{N} \gamma^\mu \nu_\mu + \text{h.c.}] + \epsilon e Z'_\mu J_{\text{em}}^\mu, \quad (8)$$

where the kinetic-mixing parameter ϵ controls the coupling of Z' to SM charged fermions and ensures that $Z' \rightarrow \mu^+ \mu^-$ proceeds with width $\Gamma(Z' \rightarrow \mu^+ \mu^-) = (\epsilon e)^2 m_{Z'}/(12\pi) \sqrt{1 - 4m_\mu^2/m_{Z'}^2} (1 + 2m_\mu^2/m_{Z'}^2)$ for $m_{Z'} > 2m_\mu$. The full free parameters of the benchmark are $(m_N, m_{Z'}, |U_{\mu 4}|^2, g_D, \epsilon)$, with the first four controlling the production cross section and the last two – through the combination $\alpha_D \epsilon^2$ – the leptonic branching of Z' .

B. Trident-like signal

At SBN energies the dominant production channel is coherent neutral current up-scattering of ν_μ into N ,

$$\nu_\mu A \longrightarrow N A, \quad (9)$$

mediated by Z' exchange between the nucleus and the ν_μ - N vertex. The cross section scales as

$$\sigma_{\nu A \rightarrow N A} \propto |U_{\mu 4}|^2 (\varepsilon g_D)^2 \frac{Z^2}{(Q^2 + m_{Z'}^2)^2} F_A^2(Q^2), \quad (10)$$

and is enhanced for sub-GeV $m_{Z'}$ through the propagator pole and the coherent Z^2 factor. Once produced, the HNL travels a macroscopic distance set by its lifetime and decays predominantly via

$$N \longrightarrow \nu Z'^{(*)} \longrightarrow \nu \mu^+ \mu^-, \quad (11)$$

where the intermediate Z' is on shell when $m_N > m_{Z'} + 2m_\mu$ and off shell otherwise. The visible final state at the LArTPC is a collinear $\mu^+ \mu^-$ pair with no associated hadronic activity beyond nuclear de-excitation – topologically identical to coherent SM trident. Discrimination is possible only through differential distributions: the dimuon invariant mass peaks at $m_{Z'}$ in the on-shell regime, the lab-frame opening angle is set by the boost $\gamma_N = E_N/m_N$ rather than by m_μ/E_ν , and the dimuon system carries an apparent “missing transverse momentum” from the escaping light neutrino in the cascade.

C. Connection to the MiniBooNE excess and existing constraints

The dark neutrino benchmark above was originally introduced as a candidate explanation of the MiniBooNE low-energy excess [22–25], in which N decays to a collimated e^+e^- pair (interpreted by the Cherenkov detector as a single electromagnetic shower) for $m_{Z'}$ slightly below $2m_\mu$. The dimuon channel is kinematically opened for $m_{Z'} > 2m_\mu$, and is a *prediction* of the same model in regions of parameter space that are not directly tested by MiniBooNE. Existing constraints [21, 31–33] leave a viable window $m_{Z'} \in [2m_\mu, 1 \text{ GeV}]$, $\varepsilon \sim 10^{-3}$, $|U_{\mu 4}|^2 \sim 10^{-7}$, that we adopt as a benchmark for SBN projections.

IV. BEYOND THE STANDARD MODEL: ON-SHELL Z' EMISSION VIA THE NEUTRINO-INDUCED INVERSE PRIMAKOFF PROCESS

A. Mechanism

The second BSM scenario considered in this work is the on-shell production of a light gauge boson Z' in the Coulomb field of an argon nucleus,

$$\nu_\mu A \longrightarrow \nu_\mu Z' A, \quad Z' \longrightarrow \mu^+ \mu^-, \quad (12)$$

with the nuclear form factor providing a quasi-real photon and the Z' emitted on shell. The reaction is the neutrino analog of the photon-induced Primakoff conversion of axion-like particles in nuclear Coulomb fields,

$\gamma A \rightarrow a A$ [26], and we will refer to it as the *neutrino-induced inverse Primakoff* process: the photon line of the standard Primakoff diagram is replaced by an active neutrino line, and the Z' is in the final state. Two complementary topologies contribute to the amplitude, depending on the underlying model:

a. (a) $L_\mu - L_\tau$ -like topology. If the Z' couples to the leptonic current of an anomaly-free $U(1)'$ symmetry such as $L_\mu - L_\tau$, the relevant interaction is

$$\mathcal{L}_{L_\mu - L_\tau} = g' Z'_\mu \left(\bar{\ell}_\mu \gamma^\mu \ell_\mu + \bar{\nu}_{\mu L} \gamma^\mu \nu_{\mu L} - (\mu \leftrightarrow \tau) \right), \quad (13)$$

so that ν_μ couples directly to Z' at tree level. In this case the dominant diagram is Z' bremsstrahlung off the outgoing neutrino line in a W -exchange trident topology, with the photon from the nucleus attaching to the internal muon propagator. The on-shell pole of Z' is reached for $(p_\nu - p_{\nu'} - q)^2 = m_{Z'}^2$, which is kinematically accessible at SBN for $m_{Z'} \lesssim E_\nu \sim 1 \text{ GeV}$.

b. (b) Kinetic-mixing topology. If Z' couples to SM charged fermions only through kinetic mixing, $\varepsilon e Z'_\mu J_\mu^\text{em}$, the leading contribution to Eq. (12) proceeds via the conversion of the coherent nuclear photon into an on-shell Z' inside the nuclear field, with the neutrino current providing the necessary momentum transfer through a Z -mediated $\nu\nu\gamma^*\gamma^*$ subprocess. The amplitude is suppressed by ε but enhanced by the coherent Z^2 on the nucleus and by the small Z' width when the resonance is reconstructed in the dimuon final state.

B. Cross section and resonance enhancement

In both topologies, integrating over the Z' propagator in the narrow-width approximation factorizes the cross section as

$$\sigma_{\nu A \rightarrow \nu \mu^+ \mu^- A}^{\text{on-shell } Z'} = \sigma_{\nu A \rightarrow \nu Z' A} \text{BR}(Z' \rightarrow \mu^+ \mu^-), \quad (14)$$

provided $m_{Z'} \gg \Gamma_{Z'}$, which is amply satisfied for the benchmark couplings considered here. The associated production cross section $\sigma_{\nu A \rightarrow \nu Z' A}$ scales coherently as Z^2 and, in the $L_\mu - L_\tau$ topology, takes the simplified form

$$\sigma_{\nu A \rightarrow \nu Z' A} \sim \frac{(g')^2 \alpha G_F^2 Z^2 E_\nu^2}{\pi^3} \mathcal{F}\left(\frac{m_{Z'}}{E_\nu}, \frac{m_\mu}{m_{Z'}}\right), \quad (15)$$

where $\mathcal{F}$ is a dimensionless function calculable analytically in the soft- Z' limit and computed numerically by NEPTUNE in general. The crucial qualitative feature is the *resonant enhancement* in the dimuon invariant mass: while SM trident populates a smooth distribution peaking near $m_{\mu\mu} \sim 2m_\mu$, the inverse Primakoff process gives a sharp peak at $m_{\mu\mu} = m_{Z'}$ with width set by detector resolution rather than $\Gamma_{Z'}$.

Placeholder for Fig. 4.

FIG. 4. Differential distributions of the $\mu^+\mu^-$ system at MicroBooNE for SM trident (gray) and for representative BSM benchmarks: a dark neutrino model with $(m_N, m_{Z'}, |U_{\mu 4}|^2) = (250 \text{ MeV}, 30 \text{ MeV}, 10^{-7})$, and an inverse Primakoff Z' benchmark with $m_{Z'} = 250 \text{ MeV}$ in $L_\mu - L_\tau$ and kinetically mixed scenarios. Left: leading-muon momentum. Center: opening angle $\theta_{\mu\mu}$. Right: transverse momentum of the dimuon system.

Placeholder for Fig. 5.

FIG. 5. Dimuon invariant mass distribution at MicroBooNE for SM trident and for representative inverse Primakoff Z' benchmarks with $m_{Z'} \in \{220, 300, 500\} \text{ MeV}$, in both the $L_\mu - L_\tau$ and kinetic-mixing topologies. The dark neutrino benchmark is overlaid for comparison. Detector smearing is applied with the resolution quoted in Ref. [16].

Placeholder for Fig. 6.

FIG. 6. Expected number of dimuon trident events per year of SBN operation at SBND, MicroBooNE, and ICARUS, broken down by contribution: SM coherent + diffractive (gray), dark neutrino cascade for the benchmark of Sec. III (blue), and inverse Primakoff Z' in the $L_\mu - L_\tau$ (red) and kinetic-mixing (orange) topologies for representative parameter points.

C. Distinguishing the two BSM benchmarks

Although both BSM scenarios produce a $\mu^+\mu^-$ final state with no associated hadronic activity, they are kinematically distinct in several respects. First, the dark neutrino cascade involves an intermediate HNL with macroscopic decay length set by $c\tau_N \propto |U_{\mu 4}|^{-2} (\varepsilon g_D)^{-2} m_N^{-5}$, so that for portions of parameter space the dimuon vertex is spatially separated from the upscattering vertex; this displaced topology is absent in the inverse Primakoff process, which is prompt. Second, the dimuon invariant mass peaks at $m_{Z'}$ for the inverse Primakoff (resonant on-shell production) but at the smaller of $(m_N - m_\nu)$ and $m_{Z'}$ for the dark neutrino model, with a broader distribution arising from the three-body N decay. Third, the angular distribution of the dimuon system retains memory of the N boost in the dark neutrino case, producing a distinctive correlation between $\theta_{\mu\mu}$ and the dimuon energy that is absent in the prompt scenario. Combining these observables – $m_{\mu\mu}$, displaced vertex, $\theta_{\mu\mu}$, $p_T^{\mu\mu}$ – provides discrimination among the two benchmarks and against the SM background.

V. COMPUTATIONAL FRAMEWORK AND SBN CONFIGURATION

The numerical results presented in Sec. VI are obtained with NEPTUNE, a Monte Carlo event generator for neutrino-induced rare processes that retains the full $2 \rightarrow n$ matrix element including nuclear form factors, lep-

ton mass effects, and arbitrary BSM mediators. For the present analysis we have implemented the SM trident amplitude following the formulation of Refs. [3, 5], with separate treatments of the coherent, diffractive, and inelastic regimes matched at $|q^2| = 1/r_p^2$, and the BSM topologies of Secs. III and IV as additional matrix-element modules sharing the same nuclear input. Phase-space integration is performed with a multi-channel adaptive sampler optimized for the soft and collinear singularities of the photon propagator and of the on-shell Z' pole, and unweighted events are written in a NuHepMC-compatible format suitable for downstream detector simulation.

We use the official SBN flux files for the on-axis BNB and the off-axis NuMI fluxes incident at SBND, MicroBooNE, and ICARUS [16], accounting for their respective baselines and angular acceptances. Detector parameters (active mass, fiducial volume, momentum and angular resolutions for muons) follow the benchmarks adopted in Ref. [4], summarized in Table I.

TABLE I. Active mass, expected exposure, and dimuon trident selection efficiencies (placeholder values) for the three SBN detectors used in this analysis.

Detector	Active mass [t]	Exposure [POT]	$\epsilon_{\mu\mu}$
SBND	112	1.0×10^{21}	TBD
MicroBooNE	85	1.3×10^{21}	TBD
ICARUS	476	1.0×10^{21}	TBD

Placeholder for Fig. 7.

FIG. 7. Projected 90% C.L. sensitivity of the SBN program in the BSM parameter spaces of the two scenarios considered in this work. Left panel: $(m_{Z'}, g')$ plane in the $L_\mu - L_\tau$ inverse Primakoff scenario, with existing constraints from CCFR [12], BaBar [36], and the muon anomalous magnetic moment overlaid. Right panel: $(m_N, |U_{\mu 4}|^2)$ plane of the dark neutrino model for fixed $(m_{Z'}, \alpha_D \varepsilon^2)$, with limits from MicroBooNE's HNL searches and from peak searches in pion and kaon decays.

VI. RESULTS

[Placeholder. This section will present NEPTUNE event yields and differential distributions for the SBN detectors, following the figure ordering above. To be written after the Monte Carlo runs are complete.]

VII. CONCLUSIONS

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